

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	01 October 2026
Team ID	SWTID-2026-8331
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Record Access Evaluation	Evaluate the requested record's field value before access Apply the configured ACL/script condition Return allow or deny decision
FR-2	Access Rule Management	Create field-value-based ACL rules Update or deactivate rules Apply rules to the relevant record/action
FR-3	Access Enforcement & Audit	Block unauthorized read/update/delete actions Record allow/deny decisions Maintain an audit trail for access checks
FR-4	Policy Testing & Validation	Test rules with sample records Validate expected allow/deny outcomes Identify invalid or conflicting rule conditions

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	ACL rules and access results should be clear and easy for administrators to understand and manage.
NFR-2	Security	Access decisions must enforce least-privilege rules and prevent unauthorized record operations.
NFR-3	Reliability	The ACL evaluation should produce consistent decisions for valid requests and handle rule failures safely.
NFR-4	Performance	Access checks should evaluate field values and rules with minimal delay before the requested operation.
NFR-5	Availability	Access-control evaluation and policy data should remain available whenever protected records are accessed.
NFR-6	Scalability	The solution should support increasing records, users, ACL rules, and access requests without major redesign.